

1. Administrative & Study Identification

Full Study Title: A Single-Dose, Randomized, Double-Blind, Placebo- and Positive-Controlled, 4-Way Crossover Study to Evaluate the Effect of Evobrutinib on the QTc (Corrected QT) Interval in Healthy Adult Participants **Short Title/Acronym:** A TQT Study of Effect of M2951 on Cardiac Repolarization / Evobrutinib Heart Safety Study **Protocol Number:** MS200527_0070 **NCT Number:** NCT07214935 **Sponsor Name:** Merck Healthcare KGaA, Darmstadt, Germany / Merck **Investigational Product:** Evobrutinib (M2951) **Phase of Development:** Phase 1 **Medical Monitor:** Merck Healthcare KGaA Clinical Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether evobrutinib has any effect on the heart's electrical activity, specifically assessing potential effects on cardiac repolarization by measuring changes in the QTc (Corrected QT) interval. **Secondary Objectives:** To validate the study's sensitivity using a moxifloxacin control, to conduct safety assessments tracking adverse events (AEs), vital signs, and clinical laboratory tests, and to measure pharmacokinetics (PK) to understand the drug's absorption and distribution.

2.2 Background & Rationale Thorough QT/QTc (TQT) studies are typically required by regulatory agencies like the FDA and EMA during clinical drug development. This Phase 1 trial is conducted early in clinical development to identify potential cardiac risks (specifically drug-induced arrhythmias associated with prolonged ventricular repolarization) before testing evobrutinib in larger patient populations.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo- and Positive-Controlled (Moxifloxacin) **Blinding:** Double-blind **Randomization:** Randomized, 4-way crossover design **Trial Status:** COMPLETED

3.2 Planned Enrollment & Duration Target \$N\$: 36 healthy adult participants **Number of Sites:** Single-center (Clinical Research Unit) **Estimated Study Start:** 08-Nov-2022 **Completion:** COMPLETED

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Overtly healthy as determined by medical evaluation, including no clinically significant abnormality identified on physical examination or laboratory evaluation and no active clinically significant disorder, condition, infection, or disease that would pose a risk to participant safety or interfere with the study evaluation, procedures, or completion. 2. Body weight within 50.0 and 100.0 kg (inclusive) and body mass index (BMI) within the range 19.0 and 30.0 kg/m² (inclusive). 3. Stable nonsmoker for at least 3 months preceding the first administration of study intervention.

4.2 Exclusion Criteria 1. History or presence of clinically relevant respiratory, gastrointestinal, renal, hepatic, hematological, lymphatic, neurological, cardiovascular, musculoskeletal, genitourinary, immunological, dermatological, connective tissue, psychiatric (due to rare risk of hallucinations, agitation, and activation of psychosis), and other diseases or disorders, and epilepsy, as determined by medical evaluation. 2. Diagnosis of hemochromatosis, Wilson's disease, alpha 1 antitrypsin deficiency, or any other chronic liver disease (including Gilbert's disease). Prior history of cholecystectomy or splenectomy, and any clinically relevant surgery within 6 months prior to the first administration of study intervention. 3. History of any malignancy. 4. History of seizures. 5. History of pharmacologically treated psychiatric disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants receive a single oral dose of one of four interventions per period: 45 mg evobrutinib (low dose), 225 mg evobrutinib (high dose), 400 mg moxifloxacin (positive control), or placebo. **Route of Administration:** Oral administration (solution/tablet) under fasted conditions. **Duration of Treatment:** Single dose administered on Day 1 of each period (4 periods total), with a minimum washout period of at least 7 days between study drug administrations.

5.2 Concomitant & Prohibited Therapy Permitted: Generally, no concomitant medications are permitted given the healthy volunteer status. **Prohibited:** Pharmacologically treated psychiatric disease medications or any other investigational therapies that might interfere with QTc assessments.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to -2 Medical Evaluation,

ECG, Demographics, Safety Labs | | **Baseline (Per Period)** | Day -1 | Admission to Clinical Research Unit (CRU), Recheck of Health Status | | **Intervention (Per Period)** | Day 1 to Day 2 | Fasted Administration, Serial ECG Monitoring, PK Blood Sampling, Vital Signs | | **Washout** | ≥ 7 Days | Clearance of previous intervention before next period | | **End of Study** | Follow-up | Safety Follow-Up Visit, Final vital signs and labs |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Placebo-corrected Change From Baseline in Corrected QT Interval by Fridericia's Formula (QTcF) for Evobrutinib. **Measurement Method:** Serial 12-lead Electrocardiograms (ECGs). A linear mixed-effects model was used to analyze the relationship between evobrutinib and MSC2729909A concentrations and Δ QTc. Based on this model, drug-induced Δ QTc and its two-sided 90% CI was predicted over the clinical concentration range and at concentrations corresponding to the observed geometric mean Cmax following administration of 45 mg and 225 mg evobrutinib. The higher geometric mean Cmax calculated based on the PK and ECG Analysis Sets was considered.

7.2 Secondary & Exploratory Endpoints * Placebo-corrected Change From Baseline in Corrected QT Interval by Fridericia's Formula (QTcF) for Moxifloxacin (to validate assay sensitivity). * Number of Participants With Treatment-Emergent Adverse Events (TEAEs). * Number of Participants With Treatment-Emergent Adverse Events (TEAEs) Based on Severity.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Severity assessed by the investigator as Mild, Moderate, or Severe. Events are classified as TEAEs if they begin or worsen after study drug dosing. **Serious Adverse Events (SAEs):** Tracked through the Safety Follow-up Visit and evaluated for relationship to study drug. **Data Monitoring Committee (DMC):** Internal safety monitoring managed by Merck in accordance with Phase 1 healthy volunteer trial standards.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: PK Analysis Set and ECG Analysis Set. **Sample Size Justification:** $N=36$ participants evaluated in a 4-way crossover design limits intra-subject variability, providing sufficient statistical power to detect small but clinically significant changes in the QTc interval. **Handling of Missing Data:** A linear mixed-effects model was used to analyze the concentration-QTc relationship to predict Δ QTc and its two-sided 90% Confidence Interval.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
Monitoring Plan: High-frequency onsite monitoring during CRU confinement (Day -1 to Day 2). **Audit Trail:** ECG core laboratory standardization and eCRF locking strictly enforced to ensure cardiac measurement validity.

11. Study Identifiers & Governance

Posting ID: 361234196472511629 **Primary ID:** MS200527_0070 **Registry Number:** NCT07214935 **Trial Status:** COMPLETED **Sponsor/Lead Organization:** Merck / Merck Healthcare KGaA **Study Title:** Evobrutinib Heart Safety Study / A TQT Study of Effect of M2951 on Cardiac Repolarization **Official Regulatory Title:** A Single-Dose, Randomized, Double-Blind, Placebo- and Positive-Controlled, 4-Way Crossover Study to Evaluate the Effect of Evobrutinib on the QTc (Corrected QT) Interval in Healthy Adult Participants

12. Clinical Framework (Cardiac Safety Specific)

Primary Condition: Healthy Volunteers (Disease Area: Healthy)
Diagnosis Threshold: Overtly healthy baseline ECG **Medical Methodology:** Thorough QT/QTc (TQT) Assessment **Optimization Target:** Evobrutinib QTc interval assessment

13. Study Procedures & Methodology

Management Pathway: 4-way crossover sequence administration
Education Protocol (HCP): TQT Protocol training, strict fasting/dosing adherence **Education Protocol (Patient):** Informed consent, CRU dietary/lifestyle rules **Quality Audit Mechanism:** Independent core ECG laboratory readings and centralized PK sampling accuracy

14. Observation & Follow-Up Schedule

Total Duration: 4 crossover periods + washout and safety follow-up
Onsite Visit Cadence: Resident Day -1 to Day 2 for each of the 4 periods
Remote Monitoring Frequency: N/A (Inpatient Phase 1 unit study)
Checklist Review Interval: Safety Follow-Up Visit at end of the study

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator				[Name]		_____			[DD-MMM-YYYY]							

Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | | Lead
Statistician | [Name] | _____ | [DD-MMM-YYYY] |